SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I Ayesha thameema banu (192524321) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ayesha Thameema Banu

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search
ii. Analyze the challenges of partial observability and dynamic environments
iii. Describe how the rover builds and updates its internal model
iv. Compare online search with uninformed and informed search strategies
v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation
ii. Explain how backtracking search works in this scenario
iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem
ii. Analyze the computational challenges of large game trees
iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
iv. Design an evaluation function and justify the factors considered
v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ARTIFICIAL INTELLIGENCE – CO2 AT2

Scenario-Based Assignment

1. AI-Powered Drone for Emergency Medicine Delivery

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

Greedy Best-First Search selects the next node based only on the heuristic value, which estimates the distance to the goal. In this scenario, the drone chooses the route that appears closest to the medicine delivery location.

Strengths:

- It makes decisions quickly.
- It uses less search effort when the heuristic is effective.
- It is useful when emergency medicines must be delivered quickly.

Weaknesses:

- It ignores the actual cost of the route already travelled.
- It may select a blocked or dangerous path.
- It does not always find the shortest or safest route.

*ii. Explain how A would handle the same situation, including how it balances cost and heuristic.**

A* Search uses both the actual path cost and the estimated remaining cost to select a route. Its evaluation function is:

$f(n) = g(n) + h(n)$

Here, g(n) represents the actual cost from the starting point to the current node, and h(n) represents the estimated cost from the current node to the destination.

The drone can use A* to consider travel distance, weather conditions and route costs. When a path becomes blocked, the system can update the route information and search for an alternative path.

iii. Analyze the impact of heuristic accuracy on both algorithms.

The accuracy of the heuristic affects the performance of both algorithms. A good heuristic guides the drone towards the destination efficiently. If the heuristic is inaccurate, Greedy Best-First Search may select a longer or unsafe route. A* can still find an optimal path when its heuristic satisfies the required conditions and the path costs are correctly represented.

iv. Discuss how dynamic changes (blocked paths) affect search performance.

Blocked roads and changing weather conditions may make the previously selected route unsuitable. The drone must update its information and search for another available route. These changes may increase computation time and delay delivery. Real-time updates help the drone avoid obstacles and select a suitable alternative.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

A* Search is the most suitable choice because it considers both the actual path cost and the estimated remaining cost. It can find an optimal route when the heuristic and cost conditions are satisfied. For safety, the drone should also consider blocked paths, weather conditions and risk levels. Real-time route updates are necessary to handle changing situations. Therefore, A* with updated route information is a suitable approach for emergency medicine delivery.

2. Smart City Traffic Signal Optimization

Formulate this as an optimization problem and explain why traditional search is inefficient.

Traffic signal optimization can be formulated as an optimization problem in which the system aims to minimize total waiting time, fuel consumption and traffic congestion.

The variables include the green-light duration, red-light duration and signal timing at each intersection. The objective function evaluates the overall traffic performance.

Traditional search methods may become inefficient because there are many possible combinations of signal timings. Traffic conditions also change continuously, making it difficult to evaluate every possible solution.

i. Explain how Hill Climbing would work in this scenario and identify its limitations.

Hill Climbing starts with an initial set of traffic signal timings and evaluates its performance. It then makes small changes to the timings and selects a neighbouring solution if it improves the objective function.

For example, the system may increase the green-light duration on a busy road to reduce vehicle waiting time.

Its limitations are that it may get stuck in a local optimum, fail to improve on a plateau, or struggle to find a good solution when the search space contains complex paths called ridges. It also depends on the initial solution.

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.

- **Local maxima:** A traffic timing arrangement may improve one intersection but fail to produce the best result for the entire city.
- **Plateaus:** Several different signal timings may produce nearly the same traffic performance, so the algorithm cannot identify a better direction easily.
- **Ridges:** Improving traffic flow may require coordinated changes at several connected intersections. Changing only one signal at a time may not produce an improvement.

These problems can prevent Hill Climbing from finding the best overall traffic arrangement.

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.

Simulated Annealing sometimes accepts a worse solution to escape a local optimum. As the search progresses, the probability of accepting worse solutions decreases. This helps the system explore more possible traffic timing arrangements.

Genetic Algorithms work with a population of possible solutions. They use selection, crossover and mutation to generate improved solutions over multiple generations. This allows the system to explore different combinations of signal timings.

Both methods can overcome some limitations of Hill Climbing by exploring a wider search space.

iv. Recommend the best approach and justify your choice based on scalability and adaptability.

A Genetic Algorithm is a suitable approach for optimizing traffic signals across a large city. It can evaluate several timing combinations and improve them over multiple generations. It is useful for handling multiple objectives and a large search space.

The system can periodically update its population using recent traffic data. However, computation time must be controlled to meet real-time requirements. Therefore, a Genetic Algorithm with regular updates and a suitable time limit can provide an effective traffic optimization solution.

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3. Autonomous Rover Exploring Mars

i. Explain why this problem requires an online search agent instead of offline search.

An online search agent is suitable because the rover does not have complete information about the Martian terrain before starting its journey. It must explore the environment, collect sensor information and make decisions while moving.

Offline search requires sufficient information about the environment in advance. Since the rover may discover unexpected rocks, craters and other hazards, an online search agent is more appropriate.

ii. Analyze the challenges of partial observability and dynamic environments.

Partial observability means that the rover cannot observe the complete environment at once. Its sensors may detect only nearby objects, and some hazards may remain hidden until it moves closer.

The rover must also handle changing environmental conditions, communication delays and limited energy. These factors make route planning difficult. Incorrect decisions may increase travel time or create safety risks.

iii. Describe how the rover builds and updates its internal model.

The rover uses cameras, sensors and other available instruments to collect information about its surroundings. It processes this information to identify obstacles, safe paths and possible sample collection locations.

As it moves, the rover updates its internal map and records newly discovered information. If an obstacle is detected, it can revise its planned route and select another safe path.

iv. Compare online search with uninformed and informed search strategies.

Uninformed search explores possible paths without using a heuristic estimate of the distance to the goal. Examples include Breadth-First Search and Depth-First Search.

Informed search uses heuristic information to guide the search towards the goal. Examples include Greedy Best-First Search and A* Search.

Online search makes decisions during exploration and updates its knowledge as new information becomes available. Unlike a search performed entirely before movement, it can adapt its actions to an incompletely known environment.

v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

An online search agent combined with heuristic-based planning is the most suitable approach for the Mars rover. It can gather information while exploring and update its internal model when new obstacles are discovered.

A* can help select efficient routes when sufficient local map information is available. The rover should also include obstacle avoidance, safe stopping conditions and energy management. This combination supports efficient exploration while reducing risks.

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4. University Examination Timetable Using CSP

Formulate the problem as a Constraint Satisfaction Problem (CSP).

A Constraint Satisfaction Problem is defined using variables, domains and constraints. The university timetable can be represented as a CSP because every examination must be assigned a suitable time and examination hall without violating the given restrictions.

i. Clearly define variables, domains, and constraints with explanation.

Variables:

Each examination is represented as a variable. For example, Mathematics, Computer Programming and Artificial Intelligence are separate variables.

Domains:

The domain contains the possible time slots and examination halls available for each examination. For example, Mathematics may be scheduled in Slot 1 or Slot 2, depending on room and faculty availability.

Constraints:

- No student should have two examinations at the same time.
- The number of examinations assigned to a hall must not exceed its capacity.
- Subjects with scheduling dependencies must follow the required order.
- Faculty availability must be considered.
- Special accommodations for students must be provided.

A valid timetable must satisfy all the hard cons